AI-Based Requirement Extraction from Audio and Video Recordings

Problem Statement

In software development, stakeholders often communicate requirements through meetings, discussions, and interviews. Manually converting these conversations into structured requirements is time-consuming and may lead to missing or ambiguous requirements.

Develop an AI-powered system that accepts an uploaded audio/video recording, converts it into text, and automatically generates:

- Functional Requirements (FR)
- Non-Functional Requirements (NFR)

The system should analyze the conversation intelligently and produce clear requirement statements.

Objective

The objective of this assignment is to:

- Understand AI-assisted Requirement Engineering
- Convert speech recordings into transcripts
- Use Generative AI/LLMs for requirement extraction
- Automatically identify and classify Functional and Non-Functional Requirements
- Generate structured software requirement outputs

Features to Implement

- Upload audio/video recording (.mp3, .wav, .mp4)
- Convert speech to text using Speech-to-Text models/APIs
- Clean and process generated transcripts
- Extract Functional Requirements (FR)
- Extract Non-Functional Requirements (NFR)
- Categorize NFRs (Security, Performance, Reliability, etc.)
- Display generated requirements in structured format
- Export generated requirements as PDF/DOCX/TXT